

- Maths is not about calculations, it's all about how math needs to be performed.
- ML algorithm cannot infer knowledge until we tell it what to learn from.

Linear Algebra

- It helps in optimization and perform operation on data like pixel rotation, shaping, sharing and all.

1. Scalar = single value.

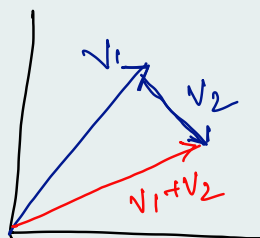
price of laptop = 50,000/- is a scalar

we can perform all arithmetic operations on it.

2. Vector = list of numbers.

→ vector Addition (not product)

— Adding 2 vectors is not simple arithmetic



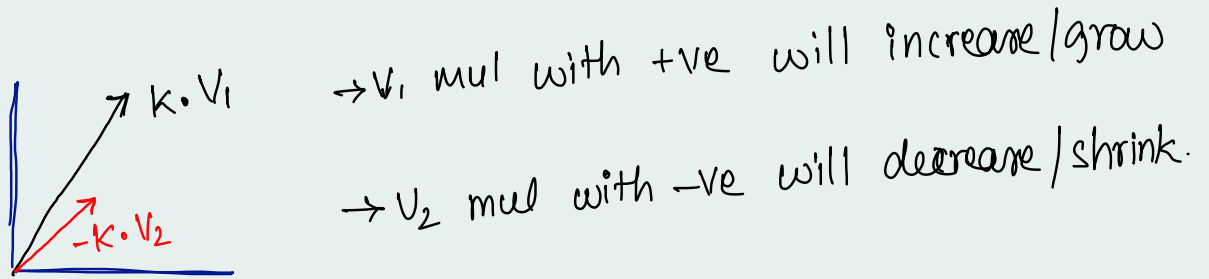
v_1 — distance

v_2 — distance

$v_1 + v_2$ — displacement

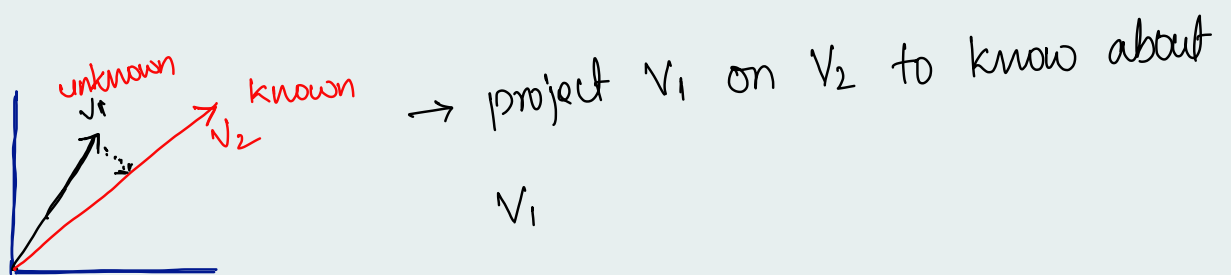
Scalar multiplication:

- when a vector is multiplied with a scalar it either grows or shrinks.



Projection:

- Projection is the shadow of a vector placed on another vector
- can help in analysing features of unknown vector



Matrices

- $\rightarrow$ we use matrices to convert our expressions of the vector form to arrays.

Suppose we have 2 equations

$$2x + 2y = 10$$

$$0x + y = 18$$

matrix form will be

$$\begin{bmatrix} 2 & 2 \\ .4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 18 \end{bmatrix}$$

Matrix Transpose:

- Interchanging rows and columns
- Transpose can be used to flip the dimensions like image rotation.

Eigen Vectors:

- Eigen vector do not change direction even if transformation is applied to them.
- we are going to use this, when there are most sensitive parts of the dataset.
- It does not change the data information, that's why we can trust on eigen vectors on our analysis.
- The values that are performing on eigen vectors are called eigen values.

Applications of Linear Algebra:

1. PCA for Dimensionality reduction helps increase quality of data
 2. Working with and applying transformations on pictures.
 3. Encoding of dataset
 4. Singular Value Decomposition (SVD)
 5. Latent semantic analysis for NLP.
 6. Optimizers in DL.
-

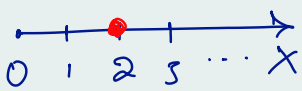
Scalar: A measurement which has only magnitude
no direction

Ex: wt, ht, age.

vector: A point which has both magnitude & direction

Ex: A bike is travelling with a speed of 50 kmph in north direction.

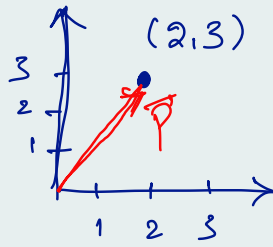
vector can represent in 1D, 2D, 3D, ..., nD.

1D repr $\rightarrow$ 

$$\vec{p} = [2]$$

column vec

2D repr $\rightarrow$



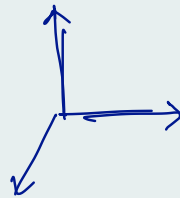
$$\vec{p} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

— vector is representing as only column vector not row

— A combination of multiple vectors are matrix.

— To repr row $\vec{p}^T = [2 \ 3]$

3D repr $\rightarrow$



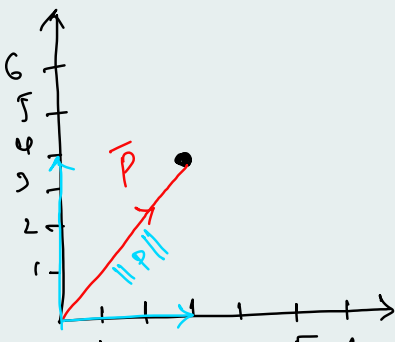
$$\vec{p} = \begin{bmatrix} 4 \\ 5 \\ 2 \end{bmatrix}$$

— In 2D plane we can repr max 3D not more

— Consider any dataset each row will repr as vector

like $\vec{p} = \begin{bmatrix} \text{RAM} \\ \text{ROM} \\ \text{price} \end{bmatrix} = \begin{bmatrix} 4 \\ 128 \\ 3000 \end{bmatrix}$

ND repr $= \vec{p} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$, $ND = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$



$\vec{p} = (3,4)$ mean vector $\vec{p}$ is moving

3 units in $\bar{x}$ -axis

4 units in $\bar{y}$ -axis

$\|p\|$ = magnitude, means the distance from origin.

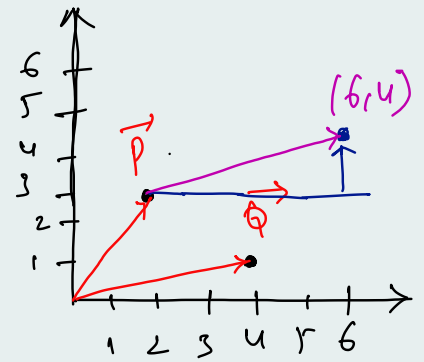
Arithmetic Operations:

1. Adding 2 vectors:

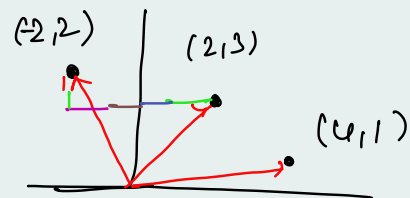
$$\vec{P} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \vec{Q} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\vec{P} + \vec{Q} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

means, $\vec{P}$ is moving 4 units in forward in x-axis
and 1 unit in forward in y-axis



$$\vec{P} - \vec{Q} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

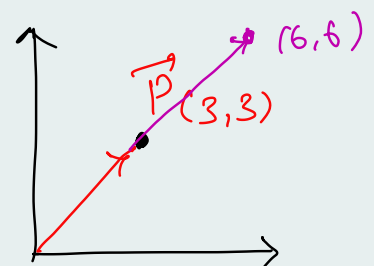


In subtraction direction got changed.

Scaling: modify the values but direction will not change.

scaling means $\vec{P} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$

$$2\vec{P} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}$$



In scaling increasing the magnitude not direction

$$\frac{1}{2} \vec{p} = \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix} \rightarrow \text{same } \downarrow \text{ mag, direction no change.}$$

Multiplication of 2 vectors:

$$\vec{p} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \vec{q} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$\vec{p} \times \vec{q} \Rightarrow$ mul rule.

$$\vec{p}^T \times \vec{q} = [2 \ 3] \begin{bmatrix} 4 \\ 1 \end{bmatrix} = [2 \times 4 + 3 \times 1] = [8 + 3] = 11$$

we are calling it as dot product

$$\boxed{\vec{p} \cdot \vec{q} = \vec{p}^T * \vec{q}} \rightarrow \text{dot product} \quad \textcircled{1} \text{ 1st repr.}$$

dot product of 2 vec = Trans multi of 2 vec.

- Dot product is also call as scalar product bcz ultimate result is scalar.

$$\vec{p} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \vec{q} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

In NID repr.

$$\vec{p} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, \quad \vec{q} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$[b_i]$

$$\vec{p} \cdot \vec{q} = [a_1, a_2, \dots, a_n] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$$\vec{p} \cdot \vec{q} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

$$\vec{p} \cdot \vec{q} = \sum_{i=1}^n a_i b_i$$

② and repr of dot prod

$a \rightarrow$ component of $\vec{p}$

$b \rightarrow$ comp of $\vec{q}$

$n \rightarrow$ no. of dimensions.

Let's have an example:

In a mobile store

no. of sales $\rightarrow \vec{S}$

no. of products $\rightarrow \vec{C}$

$$\vec{S} = \begin{bmatrix} 100 \\ 20 \\ 50 \end{bmatrix} \begin{matrix} \text{mobiles } a_1 \\ \text{headsets } a_2 \\ \text{chargers } a_3 \end{matrix}$$

$$\vec{C} = \begin{bmatrix} 5000 \\ 1000 \\ 500 \end{bmatrix} \begin{matrix} b_1 \\ b_2 \\ b_3 \end{matrix}$$

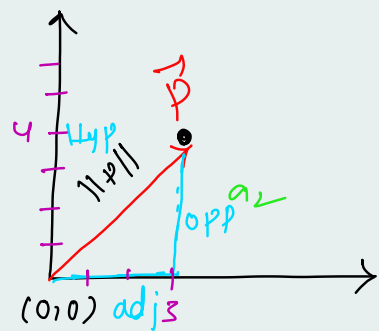
Total turn over = dot product of $(\vec{S} \times \vec{C})$

$$\vec{S} \cdot \vec{C} = \vec{S}^T \cdot \vec{C} = [100 \ 20 \ 50] \begin{bmatrix} 5000 \\ 1000 \\ 500 \end{bmatrix}$$

= 5,45,000 $\rightarrow$ total turn over.
(scalar)

Compute Magnitude of vector:

$\|\vec{p}\| \rightarrow$ length of vec from origin



As per Pythagorean Theorem

$$(\text{Hyp})^2 = (\text{opp})^2 + (\text{adj})^2$$

$$\text{Hyp} = \sqrt{\text{opp}^2 + \text{adj}^2}$$

$$\|p\| = \sqrt{4^2 + 3^2} = \sqrt{16+9} = 5.$$

Generic formula is $\|p\| = \sqrt{a_1^2 + a_2^2}$

$$3D \rightarrow \|p\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$nD \rightarrow \boxed{\|p\| = \sqrt{\sum_{i=1}^n a_i^2}}$$

n - no. of dim

$a_i \rightarrow$ comp of $\vec{p}$.

$$\|p\| = \sqrt{\sum_{i=1}^n a_i \cdot a_i} \rightarrow \text{it is looking like dot product.}$$

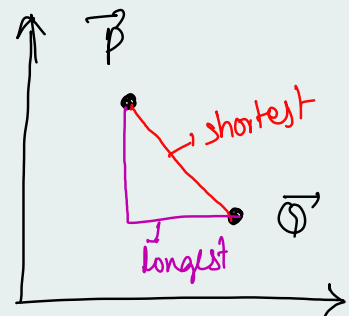
$\downarrow$
(dot prod of itself.)

$$(\sum a_i b_i)$$

$$\|p\| = \sqrt{\vec{p} \cdot \vec{p}}$$

Distance btw 2 vectors:

$$\vec{p} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad \vec{q} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$



- we can find the distance btw 2 vec in

1. shortest distance

2. longest distance

1. Shortest distance

— also called Euclidean distance

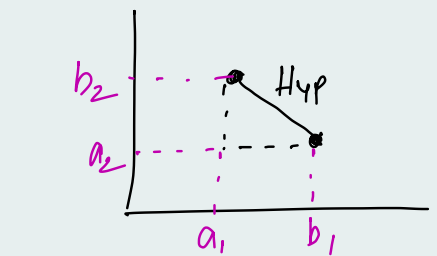
For finding shortest distance

$$\text{Hyp}^2 = \text{Side}_1^2 + \text{Side}_2^2$$

$$d^2 = \sqrt{S_1^2 + S_2^2}$$

$$\text{side}_1 = b_1 - a_1$$

$$\text{side}_2 = b_2 - a_2$$



$$\vec{P} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \vec{Q} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

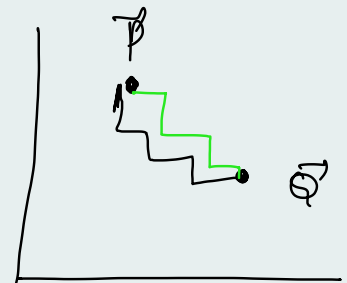
$$\therefore d = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2}$$

→ Euclidean dist.

For ND $\rightarrow d = \sqrt{\sum_{i=1}^n (b_i - a_i)^2}$

Longest distance

— also called Manhattan distance



$$d = \sum_{i=1}^n |a_i - b_i| \quad \text{will use absolute not square}$$

$$= |5 - 3| + |1 - 4| = 5 \text{ units.}$$

means '5' blocks is the longest distance btw $\vec{P}, \vec{Q}$.

— There is one more distance which combined both is Minkowski distance.

$$d(\vec{P}, \vec{Q}) = \left[\left(\sum_{i=1}^n |b_i - a_i| \right)^p \right]^{1/p}$$

if $p=1 \rightarrow$ Manhattan dist

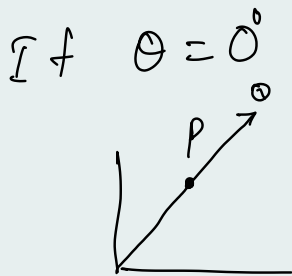
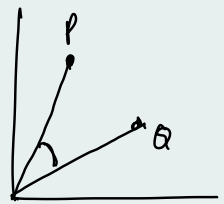
$p=2 \rightarrow$ Euclidean dist

Dot product repr:

$$1. \vec{P} \cdot \vec{Q} = \vec{P}^T \times \vec{Q}$$

$$2. \vec{P} \cdot \vec{Q} = \sum_{i=1}^n a_i b_i$$

$$3. \vec{P} \cdot \vec{Q} = \underbrace{\|\vec{P}\|}_{\text{magn}} * \underbrace{\|\vec{Q}\| * \cos \theta}_{\text{direction}}$$

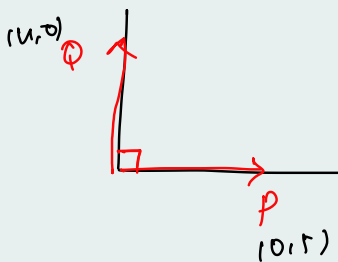


$$\cos 0 = 1$$

$$\vec{P} \cdot \vec{Q} = \|\vec{P}\| \cdot \|\vec{Q}\|$$

which is directing in same direction

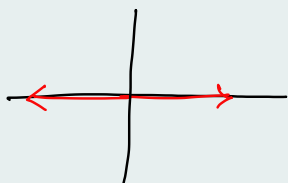
if $\theta = 90^\circ$



$$\vec{P} \perp \vec{Q} = 0 \quad \left\{ \begin{array}{l} \text{dot prod is 0} \\ \cos 90 = 0 \end{array} \right.$$

$$\vec{P} = (0, 1), \vec{Q} = (1, 0) \Rightarrow \vec{P} \cdot \vec{Q} = 0$$

if $\theta = 180^\circ$

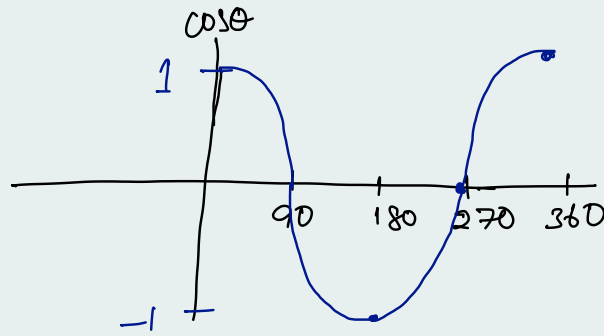


$$\cos 180 = -1$$

$$\vec{p} \cdot \vec{q} = -\|\vec{p}\| \cdot \|\vec{q}\|$$

$$\cos \theta = [-1, +1]$$

cos θ graph:



if

$$\theta = 0-90 \rightarrow \cos \theta = 0 \text{ to } 1$$

$$\theta = 90-180 \rightarrow \cos \theta = -1 \text{ to } 0$$

⋮

Cosine Similarity:

$$\vec{p} \cdot \vec{q} = \|\vec{p}\| \cdot \|\vec{q}\| \cdot \cos \theta$$

used for find the similarity btw 2 vec.

- If the angle btw 2 vec is less $\rightarrow$ more similar
- angle btw 2 vec is high $\rightarrow$ not similar.

$$\cos \theta = \frac{\vec{p} \cdot \vec{q}}{\|\vec{p}\| \cdot \|\vec{q}\|}$$

$$\cos \theta = \frac{\sum_{i=1}^n a_i b_i}{\sqrt{\sum_{i=1}^n a_i^2} \cdot \sqrt{\sum_{i=1}^n b_i^2}}$$

